



BAULKHAM HILLS HIGH SCHOOL

JULY 2013
YEAR 11 TASK 2

Mathematics

General Instructions

- Working time – 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 6 to 13
- Marks may be deducted for careless or badly arranged work

Total marks – 55

Exam consists of 4 pages.

This paper consists of TWO sections.

Section 1 – Page 2

Questions 1-5 (5 marks)

- Attempt Question 1-5

Section II – Pages 3-4 (50 marks)

- Attempt questions 6 to 13

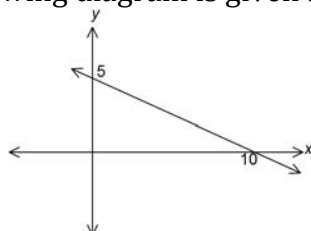
Section I - 5 marks
Attempt questions 1-5

Use the multiple choice answer sheet for question 1-5

1. Evaluate $\lim_{x \rightarrow 5} \frac{2x^2 - 11x + 5}{x^2 - 25}$

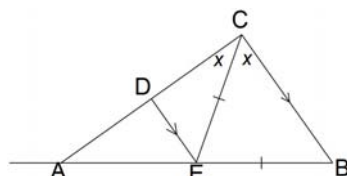
- (A) 0 (B) $\frac{9}{25}$ (C) $\frac{9}{10}$ (D) not defined

2. Equation of the line in the following diagram is given by



- (A) $y = \frac{1}{2}x + 5$ (B) $y = x + 5$
 (C) $y = -\frac{1}{2}x + 5$ (D) $y = -\frac{1}{2}x + 10$

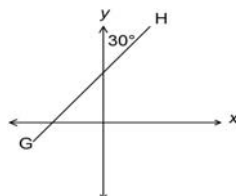
3.



In the diagram, $EB = EC$, $\angle BCE = \angle DCE$ and $BC \parallel ED$. $\triangle CDE$

- (A) is equilateral (B) is isosceles
 (C) is scalene (D) cannot be determined

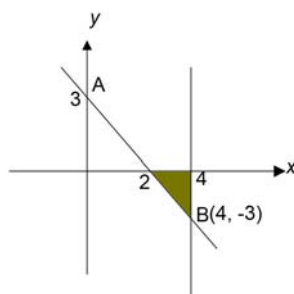
4.



The line GH makes an angle of 30° with the y -axis, as shown in the diagram. What is the gradient of GH

- (A) $\sqrt{3}$ (B) $\frac{1}{2}$
 (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

5.



The three inequalities to describe the shaded region given above are

- (A) $y \leq 0, x \geq 2, 3x - 2y + 6 \leq 0$ (B) $y \leq 0, x \leq 4, 3x - 2y - 6 \leq 0$
 (C) $y \leq 0, x \geq 2, 3x - 2y - 6 \geq 0$ (D) $y \leq 0, x \leq 4, 3x - 2y + 6 \geq 0$

Section II – Extended Response

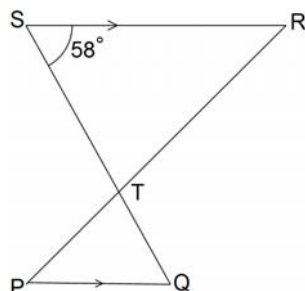
Attempt questions 6-13.

Answer each question on a SEPARATE PAGE. Clearly indicate question number.

Each piece of paper must show your name.

All necessary working should be shown in every question.

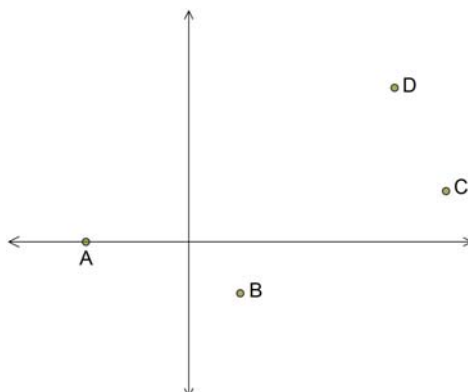
6.



Given $SR \parallel PQ$ and that SQ, RP are straight lines, and $\angle RST = 58^\circ$, $\angle PTQ = 53^\circ$, find the size of $\angle TPQ$. (without giving reason)

1

7.



Consider the quadrilateral $ABCD$ with $A(-2, 0)$, $B(1, -1)$, $C(5, 1)$ and $D(4, 3)$.
Copy the diagram with all the information given.

- (i) Find the lengths of AD and BC .
- (ii) Find the gradient of AD .
- (iii) Show that the equation of the line passing through A and D is given by $x - 2y + 2 = 0$.
- (iv) Show that $AD \parallel BC$.
- (v) What type of quadrilateral is $ABCD$?
- (vi) Calculate the perpendicular distance of B from the line AD .
- (vii) Calculate the area of $ABCD$.

3

1

2

2

1

2

2

8. Differentiate the following

(i) $y = 5x^3 + \frac{1}{x}$

(ii) $y = \frac{3x^2 + 10x}{x}$

(iii) $y = (3x + 7)(1 - x)^3$

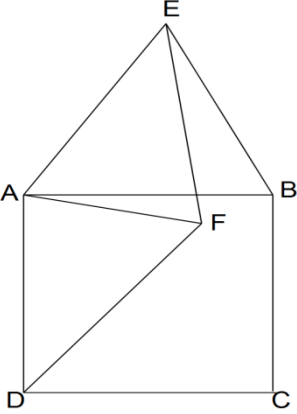
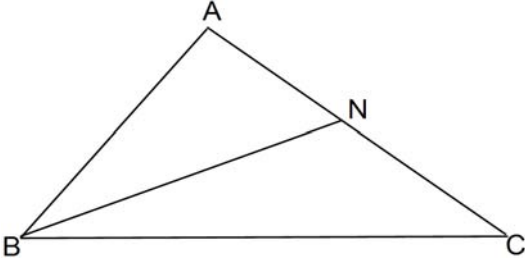
(iv) $y = \frac{5x^2 - 2}{2x + 1}$

2

2

2

2

9.	Find the derivative of the function given by $y = x^2 + 2x$ using first principles.	3
10.	Find the equation of the line which is passing through the intersection of $x - 3y + 5 = 0$ and $x + 2y = 0$ and the point $(3, 1)$.	3
11.	(i) Show that the equation of the normal to the curve $y = \frac{1}{\sqrt{x^2-3}}$ at $(2,1)$ is given by $x - 2y = 0$. (ii) Find the coordinates of the point Q where the normal meets the curve again.	2 3
12.	 In the diagram $ABCD$ is a square and AEB is an equilateral triangle. EF bisect $\angle AEB$. $\angle EAF = \angle DAF$. Copy the diagram and label the given information. (i) Show that $\angle BAF = 15^\circ$. (ii) Prove that $EF = DF$	3 4
13.	ABC is a triangle and N is point on AC. $\angle ABN = \angle CBN = \angle BCN = \theta$. $BC = 2a$, $CA = b$, $AB = c$. $BN = d$.  (i) Copy the diagram, labelling all of the given information. (ii) Explain why $CN = d$ (iii) Prove that $\angle ANB = \angle ABC$ (iv) Prove that $\triangle ABN$ is similar to $\triangle ACB$. (v) Show that $2ac = bd$ (vi) Hence show that $c^2 = b^2 - 2ac$	1 1 2 2 2 2

~ End of Exam ~

M.C

1) C 2) C 3) B 4) A

$$\underline{26} \quad \angle TPO = 69 \quad \text{--- (1)}$$

$$\underline{27} \quad \text{(i)} \quad AD = \sqrt{(4+2)^2 + 3^2} \quad \text{--- (1)}$$

$$= \sqrt{36+9} = 4.5 \quad \text{--- (1)}$$

$$\text{or } 3\sqrt{5}$$

$$\text{(ii)} \quad BC = \sqrt{(5-1)^2 + 2^2}$$

$$= \sqrt{20} = 2\sqrt{5} \quad \text{--- (1)}$$

$$\text{(iii)} \quad m_{AD} = \frac{3-0}{4+2} = \frac{3}{6} = \frac{1}{2} \quad \text{--- (1)}$$

$$\text{(iv)} \quad y-0 = \frac{1}{2}(x+2) \quad \text{--- (1)}$$

$$\left. \begin{array}{l} 2y = x+2 \\ x-2y+2=0 \end{array} \right\} \quad \text{--- (1)}$$

$$\text{(v)} \quad m_{BC} = \frac{1+1}{5-1} = \frac{2}{4} = \frac{1}{2} \quad \text{--- (1)}$$

$$\therefore m_{AD} = m_{BC} \quad \text{--- (1)}$$

Hence $AD \parallel BC$

(vi) Trapezium --- (1)

$$\text{vii)} \quad D = \frac{|1 \times 1 - 2 \times -1 + 2|}{\sqrt{1+2^2}} \quad \text{--- (1)}$$

$$= \frac{5}{\sqrt{5}} = \sqrt{5} \quad \text{--- (1)}$$

$$\text{vii)} \quad A_r = \frac{1}{2} \times h \times (a+b)$$

$$= \frac{1}{2} \times \sqrt{5} \times (3\sqrt{5} + 2\sqrt{5})$$

$$= \frac{1}{2} \times \sqrt{5} \times (5\sqrt{5})$$

$$= \frac{25}{2} = 12.5 \text{ unit}^2 \quad \text{with working \& correct ans.} \quad \text{(2)}$$

$$\underline{Q8} \quad \text{(i)} \quad y' = 15x^2 - \frac{1}{x^2} \quad \text{--- (1)}$$

$$\text{(ii)} \quad y = 3x + 10$$

$$y' = 3 \quad \text{--- (2)}$$

$$\text{(iii)} \quad y' = 3(1-x)^3 + (3x-7) \times 3(1-x)^2 \times -1 \quad \text{--- (1)}$$

$$= 3(1-x)^3 - 3(3x-7)(1-x)^2 \quad \text{--- (1)}$$

$$= 3(1-x)^3 [1-x - 3x+7]$$

$$= 3(1-x)^3 (8-4x) = 12(1-x)^3 (2-x)$$

$$\text{(iv)} \quad y' = \frac{(2x+1)(10x) - (5x^2-2)(2)}{(2x+1)^2} \quad \text{--- (1)}$$

$$= \frac{20x^2 + 10x - 10x^2 + 4}{(2x+1)^2}$$

$$= \frac{10x^2 + 10x + 4}{(2x+1)^2}$$

$$\underline{Q9} \quad f(x) = y = x^2 + 2x \quad \text{--- (1)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 2(x+h) - x^2 - 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2xh + x^2 + 2x + 2h - x^2 - 2x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + 2xh + 2h}{h} \quad \text{--- (1)}$$

$$= \lim_{h \rightarrow 0} h + 2x + 2$$

$$f'(x) = 2x + 2$$

Q10 $x - 3y + 5 = 0$
 $x + 2y = 0$

$-5y + 5 = 0$

$y = 1$
 $x = -2$ } — (1)

∴ eq. of the line passing through $(-2, 1)$ & $(3, 1)$

$m = \frac{1-1}{3+2} = 0$ — (1)

∴ eq. is $y - 1 = 0(x + 2)$ — (1)
 $y = 1$

OR $x - 3y + 5 + k(x + 2y) = 0$ — (1)
 $3 - 3k + 5 + k(3 + 2) = 0$
 $k = -1$ — (1)

∴ eq. is $x - 3y + 5 - 1(x + 2y) = 0$
 $x - 3y + 5 - x - 2y = 0$
 $-5y = -5$
 $y = 1$ } (1)

iii (i) $y = (x^2 - 3)^{-1/2}$
 $y' = -\frac{1}{2}(x^2 - 3)^{-3/2} \cdot 2x$
 $= -x(x^2 - 3)^{-3/2}$ — (1)

m_T at $x = 2$
 $m_T = -2(4 - 3)^{-3/2}$
 $= -2$

$m_N = \frac{1}{2}$ — (1)

eq. of Normal $y - 1 = \frac{1}{2}(x - 2)$
 $\Rightarrow 2y - 2 = x - 2$
 $x - 2y = 0$

i) Pt. of int. of the Normal to the curve $\left(\frac{x}{2}\right)^2 = \left(\frac{1}{\sqrt{x^2 - 3}}\right)^2$ — (1)

$\frac{x^2}{2} = \frac{1}{x^2 - 3}$

$x^4 - 3x^2 = 4$

Sub $x^2 = t$

$t^2 - 3t - 4 = 0$ — (1)

$(t - 4)(t + 1) = 0$

$t = 4, t = -1$

$x^2 = 4, x^2 = -1$

$x = \pm 2$ not possible — (1)

$x = 2, y = 1$ or $x = -2, y = 1$

$(-2, 1)$ — (1)

OR Does not touch the curve anywhere.

Q12 In dia ABCD is a sq & AEB is an equilateral Δ . The line EF bisects $\angle AEB$ ~~and~~
 $\angle EAF = \angle DAF$

(i) Let $\angle BAF = x$
 $\angle EAB = 60^\circ$ (all $\angle$ s of equilateral Δ)
 $\therefore \angle EAF = 60^\circ + x$ — (1)
 $\angle DAB = 90^\circ$ (ABCD is a sq)
 $\angle DAF = 90^\circ - x$ — (1)

$90^\circ - x = 60^\circ + x$ (given $\angle DAF = \angle EAF$)
 $2x = 30^\circ$
 $x = 15^\circ$

(ii) In Δ 's DAF & EAF

① $AD = AB$ (sides of a sq)

$AE = AB$ (sides of equilateral Δ)

$\therefore AD = AE$ — (1)

$\angle DAF = \angle EAF$ given

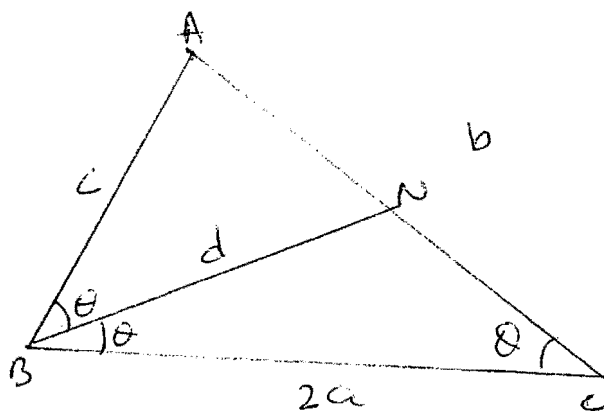
AF is common

$\therefore \Delta AEF \cong \Delta ADF$ (SAS) — (1)

$\therefore EF = DF$ matching sides

$\Delta \cong \Delta$ — (1)

Q13.
= (i)



- (i) $BN = CN$ (sides opp equal $\angle$ s) — (1)
 $\therefore BN = CN = d$ — (1)

- (ii) $\angle BNC = 180 - 2\theta$ ($\angle$ sum of Δ) — (1)
 $\therefore \angle ANB = 2\theta$ (st line $\angle$) — (1)
 Also $\angle ABC = \theta + \theta = 2\theta$ — (1)
 Hence $\angle ANB = \angle ABC$ — given

- (iv) In $\triangle ABN$ & $\triangle ACB$
 $\angle BAC$ common — (1)
 $\angle ANB = \angle ABC$ (proved above)
 $\therefore \triangle ABN \sim \triangle ACB$ (all matching $\angle$ s are =) — (1)

- v) $\frac{BN}{BC} = \frac{AN}{AB} = \frac{AB}{AC}$
 (matching sides are in the same ratio in $\sim \Delta$ s) — (1)

$$\frac{d}{2a} = \frac{b-d}{c} = \frac{c}{b} \quad \text{--- (1)}$$

$$\frac{d}{2a} = \frac{c}{b}$$

$$bd = 2ac.$$

- (vi) $\frac{b-d}{c} = \frac{c}{b}$ — (1)
 $c^2 = b^2 - bd$ — (1)
 but $bd = 2ac$ (from above)
 $\therefore c^2 = b^2 - 2ac.$